

product information
as a nutraceutical.

ZONPea™

Palmitoylethanolamide



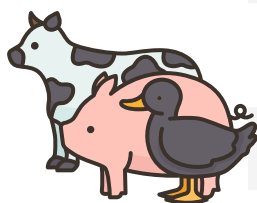
PALMITOYLETHANOLAMIDE

Palmitoylethanolamide (ZONPea) is an endocannabinoid (eCB)-like bioactive lipid mediator, belonging to the N-acyl-ethanolamine (NAE) fatty acid amide family. First isolated from soybean lecithin and egg yolk in 1957, PEA was initially identified to have anti-inflammatory and anti-allergic effects.

Some Plant and animal sources of ZONPea



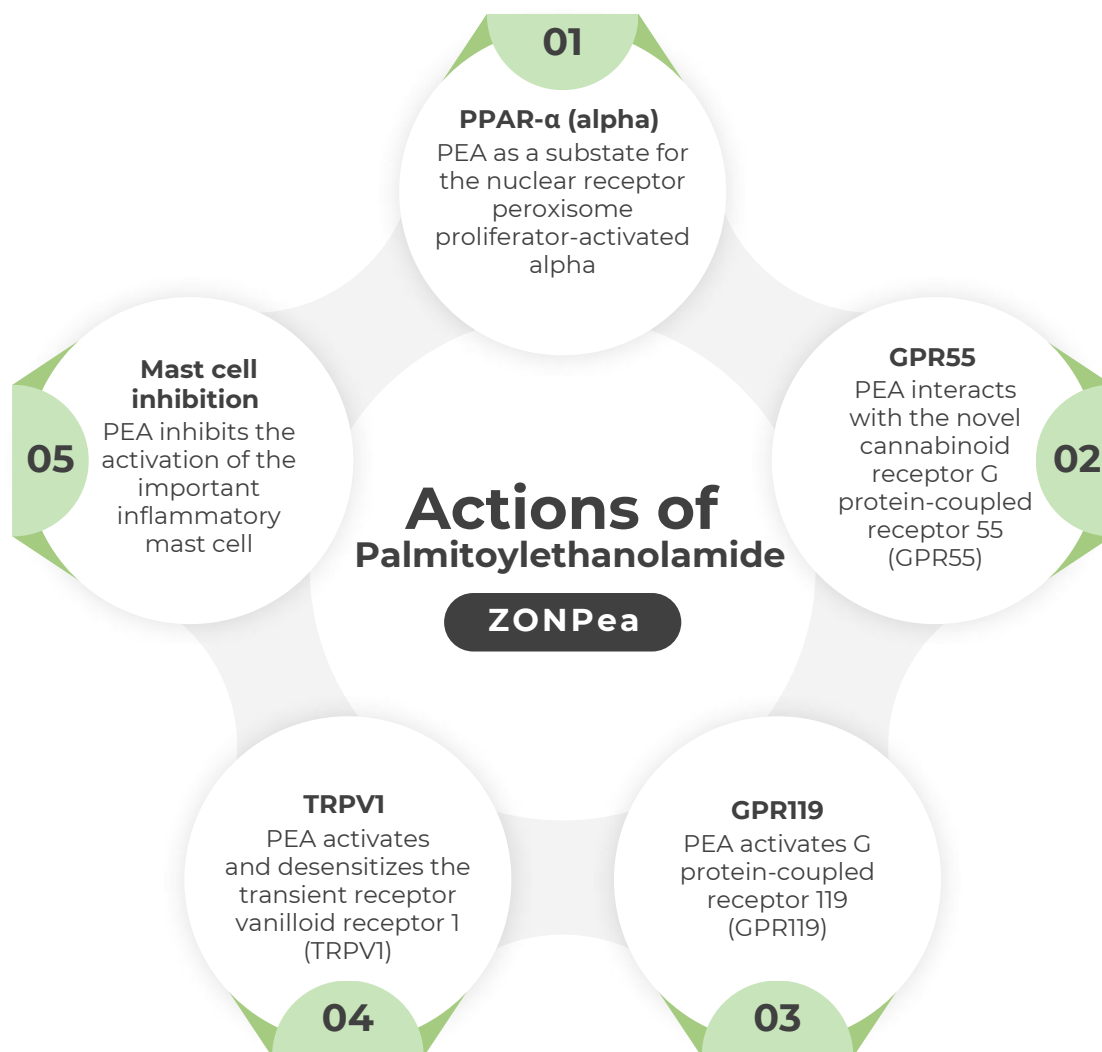
Food Source	ng/g f.w.
Soy lecithin	950,000
Soybean (<i>Glycine max</i>)	6700
Green coffee (depending on the variety)	2830-11,940
Raw peanuts	~7770
Roasted coffee	7200
Peanuts (<i>Arachis hypogaea</i>)	3730
Alfalfa (<i>Medicago sativa</i>)	1150
Tuna fish	~120 *
Chicken	~120 *
Eggs	~ 100 *
Beef	~ 60 *
Codfish	~ 60 *



f.w. = fresh-weight; * = the concentration is expressed as nanograms per gram on a dry-weight basis; ~ indicates the value has been extrapolated from figures.

MECHANISM OF ACTION

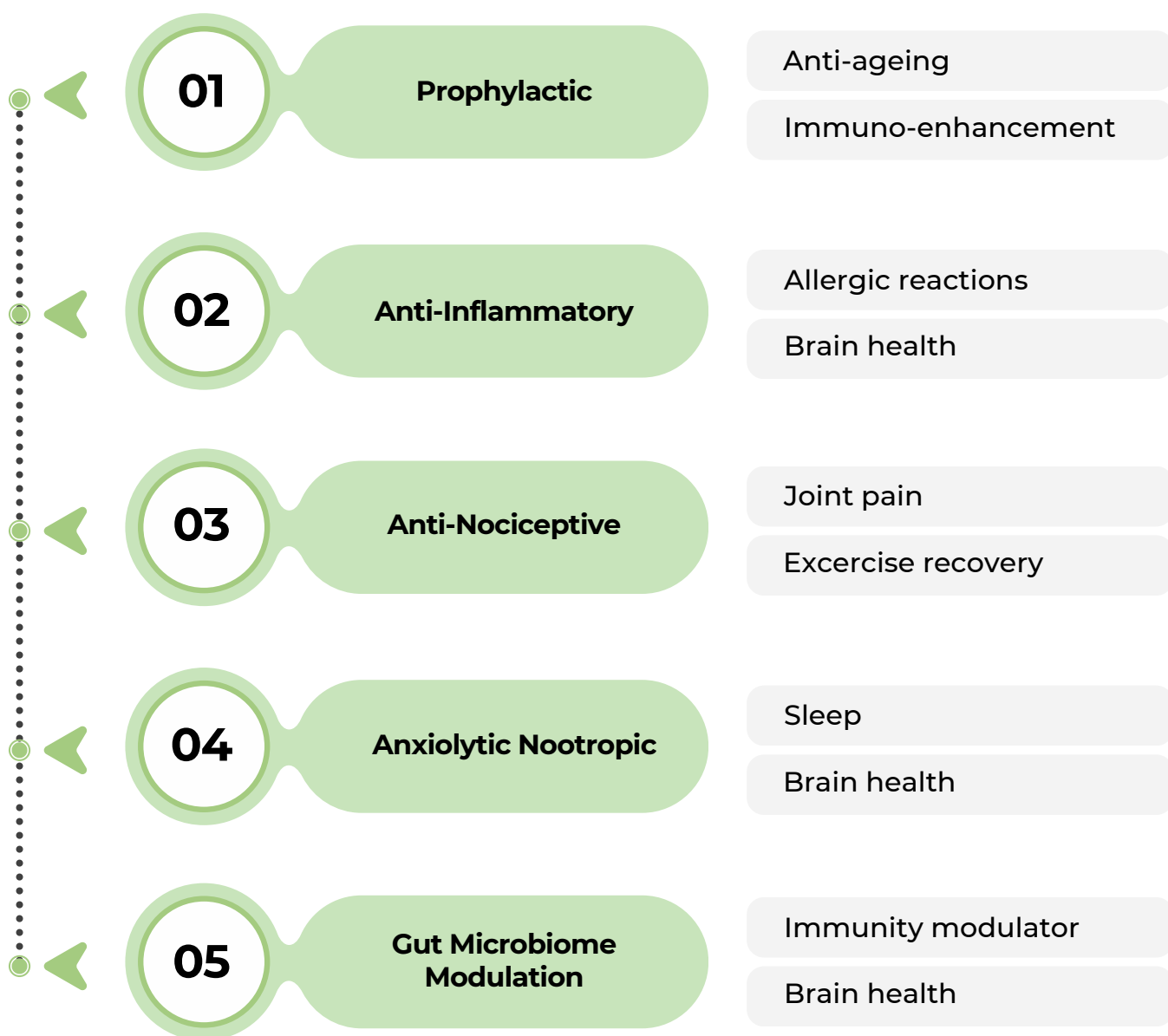
Palmitoylethanolamide (ZONPea) is characterized as autacoid local inflammation antagonism (ALIA). Abundantly found in central nervous system and are produced in demand by several cell types including mast cells, astrocytes, and microglia.



ZONPea's multiple receptor mechanisms of action generate therapeutic benefits in many disorders, including allergic reactions, influenza, common cold, chronic pain, joint pain, psychopathologies and neurodegeneration. Moreover, it indirectly activates cannabinoid receptors 1 and 2 (CB1 and CB2) through inhibiting the degradation of the endocannabinoid, anandamide (AEA), a phenomenon known as the 'entourage effect'.

ZONPea's PLEIOTROPIC EFFECTS

ZONPea is thought to be produced as a pro-homeostatic protective response to cellular injury and is usually up-regulated in disease states. ZONPea's multifaceted effects are due to its unique mechanisms of action that affect multiple pathways at different sites.



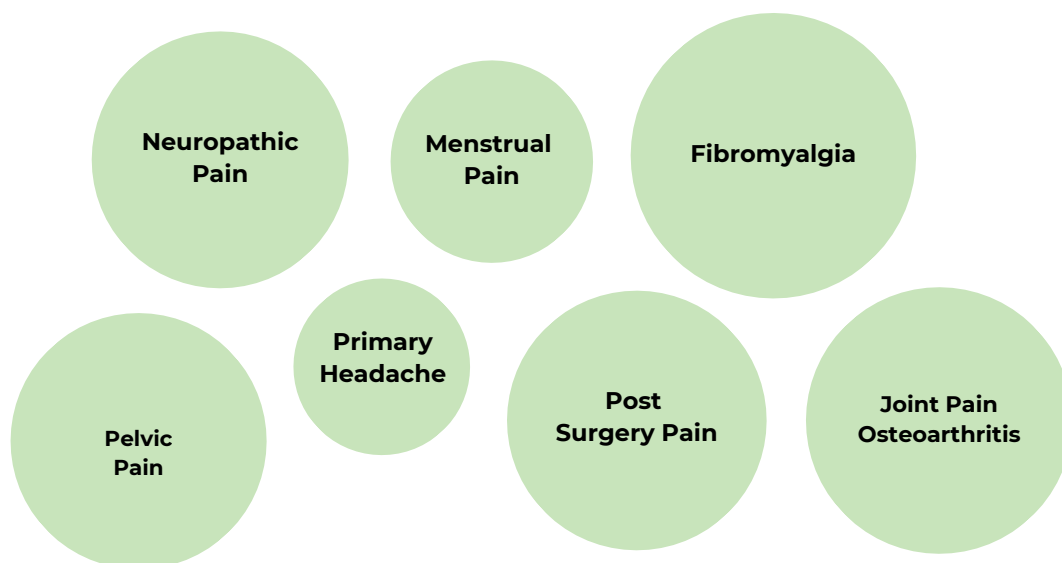
PAIN MANAGEMENT

The modern lifestyle, diet, and the ageing process create imbalances which drive chronic inflammation. The synthetic anti-inflammatory drugs have applications in the short-term management of chronic inflammation but are too toxic for long-term and/or preventive use at the public health level.



Chronic pain affects many people in the Western world, constituting an enormous burden for the individuals and society. According to the American Pain Society, the prevalence of chronic pain in the US is

35.5%



Among anti-inflammatory and pro-resolving lipid mediators, palmitoylethanolamide (ZONPea) has been reported to down-modulate mast cell activation and to control glial cell behaviors.

Numerous preclinical studies demonstrate the ability of ZONPea to reduce inflammation and pain induced by various acute stimuli. The effect of ZONPea administration is dose-dependent. The anti-inflammatory and analgesic effect of ZONPea have been confirmed in models of chronic inflammation.

NEURO PROTECTION

ZONPea's neuroprotective effects are due to its ability to modify MC, microglia and astrocyte activation. It enhances microglial migration without promoting activation, thereby increasing resistance to infection without measurable pro-inflammatory effects. ZONPea's inhibition of pro-inflammatory cytokines likely contributes to its ability to prevent cortical spreading depression in pre-clinical models and its therapeutic effects in migraine as well.

ZONPea Neuroprotective applications

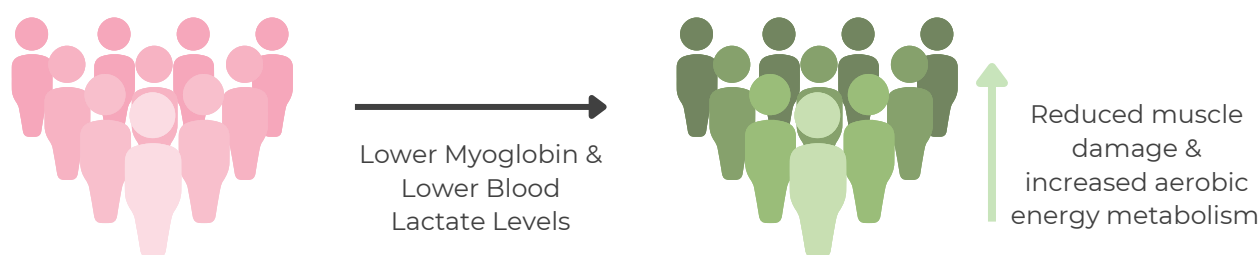


Moreover, ZONPea supports the body's natural regulatory processes by balancing inflammation in the nervous system. This mechanism can help to maintain cellular health and mitigate damage, particularly in neurodegenerative conditions. While ongoing research is needed, ZONPea shows promise as a supportive element in the management of neuroinflammation and associated disorders.

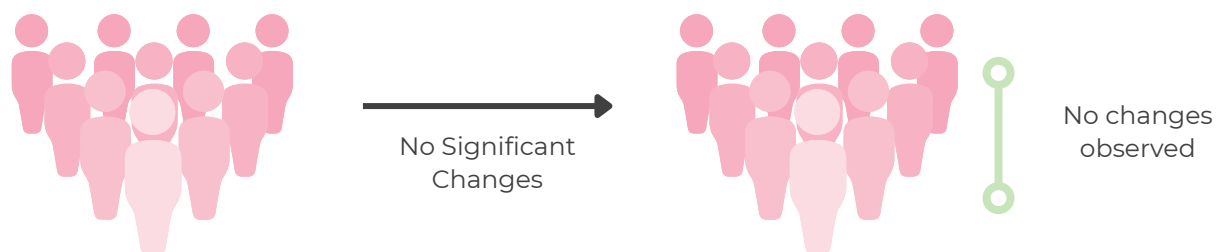
EXERCISE RECOVERY

Injuries are a part of life, and a primary reason why people stop exercising. Rest and recuperation are often indicated but safe analgesic/anti-inflammatory support is also desirable, especially in the ageing population or occasional athletes.

Case 1: 176.5mg of PEA (Liquid Form)



Case 2: Placebo

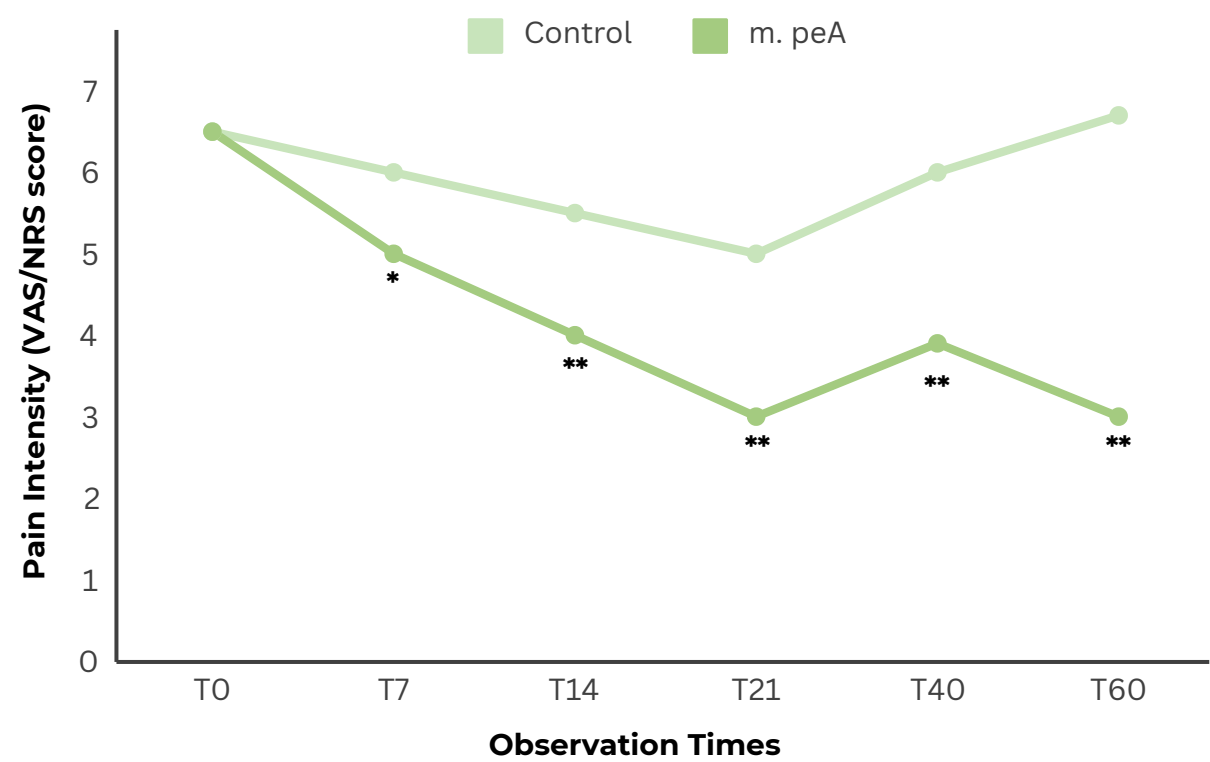


Source: Clayton, P.; Hill, M.; Bogoda, N.; Subah, S.; Venkatesh, R. Palmitoylethanolamide: A Natural Compound for Health Management. *Int. J. Mol. Sci.* 2021, 22, 5305. <https://doi.org/10.3390/ijms22105305>

The above diagram demonstrates a clinical trial assessed the impact of PEA on recovery from muscle-damaging exercise. The findings associated with enhanced recovery and the ability to maintain higher exercise intensities for longer. The results are in line with ZONPea's ability to reduce muscle protein breakdown as shown by a significant increase in protein kinase B (Akt) phosphorylation, a kinase known to induce protein synthesis, and the ability to clear myoglobin from the circulation. The overall results indicate that ZONPea may enhance concurrent training adaptations after exercise.

FSMP (FOOD FOR SPECIAL MEDICAL PURPOSES)

FSMPs are formulated to meet the specific dietary needs of patients with certain medical conditions. ZONPea-based FSMPs can be integrated into a patient's diet to complement existing treatments.



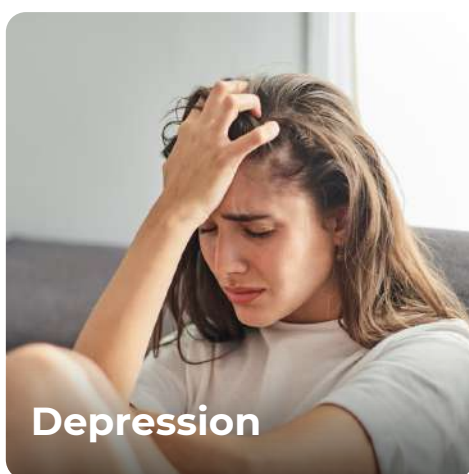
	T0	T1	T2	T3	T4	T5
Δ^*	-0.16	0.46	1.21	1.65	3.34	3.52
p	0.9738	0.0492	<0.001	<0.001	<0.001	<0.001

Δ^* = Efficacy estimate **m. PEA / u.m. PEA** groups vs control group

Above figure demonstrates changes in pain intensity in patients treated with PEA and control groups at different observation times. Values are expressed as mean $\pm$ SEM

MENTAL HEALTH AND MOOD SUPPORT

ZONPea levels are increased by acute psychosocial stress, presumably as a protective response to aversive and/or hazardous situations and a modifier of the induced Cell Danger Response (CDR). ZONPea's antidepressant, anxiolytic and analgesic actions are potentially also involved here. ZONPea's ability to enhance neurogenesis and facilitate synaptic plasticity likely plays a role in improving behavior and cognition.



ZONPea's ability to target neuro-inflammation, pain, depression, anxiety and at the same time support neurogenesis and synaptic pruning makes it a viable therapeutic aid for brain disorders.

ZONPea

Palmitoylethanolamide (PEA) crafted by HENI branded as ZONPea manufactured in GMP-certified facilities to ensure the highest quality and consistency.

Our ZONPea meets international regulations and standards. We have rigorous testing by independent labs to confirm purity and potency.

FSSAI Licensed Product for the use in Food and Nutraceutical, FSMP (Food for special medical purposes) applications.

Shelf Life: Palmitoylethanolamide (ZONPea) has a shelf life of at least 30 months from production date.

Storage conditions: Store in cool and dry place, protected from sun light.

Pack Size: 25Kg (Packed in double LDPE inner bags, further packed in an export worthy, new HDPE drum)

Appearance: White crystalline powder.

Assay by HPLC (%w/w)(on anhydrous basis) : Min 99.0%.

Typical average particle size, D(90) : 20-30 microns

USED IN



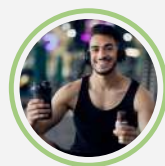
Pharmaceutical Industry

Incorporated in the formulation of medications for therapeutic health benefits.



Food & Nutraceuticals

Used in the formulation of dietary supplements for overall wellness.



Sports Nutrition

Featured in the formulation of supplements to enhance athletic performance.



Cosmetics and Personal Care

Included in the formulation of skincare products for healthier skin.

Compatible for capsules, tablets, powders, and sachets to meet diverse needs.

OUR EXPERTISE

Customization for your unique formulation

TECHNICAL SPECIFICATIONS OF ZONPea.

TESTS	REQUIREMENTS
Description	White Crystalline powder
Identification	
• By FTIR	The Infrared absorption spectrum of the sample should be concordant with the spectrum obtain from standard
• By HPLC	The retention time of the major peak in the chromatogram of the assay preparation should correspond to that in the chromatogram of the standard preparation, as obtained in the assay.
Aroma	Characteristic
Solubility	Practically insoluble in water
Assay by HPLC (%w/w) (on anhydrous basis)	Not less than 99.0 %
Water content by KF (%w/w)	Not more than 0.5 %
Melting Point (°C)	Between 98.0 to 102.0
Ash (%w/w)	Not more than 0.10 %
Particle size D (90)	90 % less than 60 µm
Bulk density	
a. Loose	Not less than 0.2 g/cc
b. Tapped	Not less than 0.3 g/cc
pH	Between 6.0 to 7.0
Related Compound : (By HPLC)	
a. Palmitic acid	Not more than 0.5%
b. Stearoylethanolamide	Not more than 0.2%
c. Myristoylethanolamide	Not more than 0.2%
d. Individual Maximum impurity	Not more than 0.1%
e. Total Impurities	Not more than 1.0%
Total Heavy Metals	Not more than 10.0 ppm

Production in a GMP and TUV Nord approved ISO9001 : 2015 environment, under brand name ZONPea.





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Coming Soon...

PEA for application in derma-
pharma, personal care such as

Shampoos

Sunscreen

Hair Care

Oral Care

Skin Care

To know more get in touch with us!